HOW MUCH HUMAN WORK IS BEHIND AI-ASSISTED SOFTWARE?
LABOUR BOUNDS AND ESTIMATION FROM WORK TELEMETRY

Abstract. Software-work telemetry records events whose relationship to human time can change under AI assistance. We determine which labour totals, measured in allocated person-hours, are consistent with specified observations. We report elapsed time to acceptance and output retained at follow-up separately. Observed interval totals or justified lower and upper bounds, together with each person's available time, constrain how much labour could have occurred within a target window. Splitting time at every observation endpoint and target-window boundary gives a finite linear programme with the same possible labour totals, even when observations overlap or tasks share a person's time. Minimising and maximising the target total gives its exact bounds; every total between them is possible under the same constraints. Adding bounds calculated separately for each interval can include totals that no joint allocation can produce. For a probability sample, we add differences between observed and predicted labour, weighted by inverse inclusion probabilities, to the total of predictions fixed before sampling. Under the stated sampling and measurement assumptions, this estimate averages to true labour over repeated samples. Its variance can be higher or lower than that of the estimator without predictions. We supply analytical proofs and exact finite computational checks using constructed data. The proposed measurements still require empirical validation.

1. What the observations determine

Suppose a team records three person-hours of work during a four-hour interval. How much work occurred during its first two hours? If one person did all the work, at least one person-hour must fall in the first half, because at most two can fall in the second. At most two person-hours can fall in the first half. The person could work one hour in the first half and two in the second, or two in the first and one in the second. The recorded total therefore does not tell us exactly how much work occurred in the first two hours. Commit times alone may not even determine total labour: the same event sequence can arise from different amounts of human activity.

Labour is human time allocated to the included work, measured in person-hours. For AI-assisted software work, we count person-hours from all contributing roles and report elapsed time to first acceptance and reference output retained at follow-up separately. The accounting includes unsuccessful work and post-acceptance repair. At any time, the fractions of a person's time allocated to different tasks and roles must sum to at most one. Our linear programme gives the minimum and maximum labour totals consistent with the observed interval totals or justified bounds and each person's available time. Every total between those endpoints is possible under the same constraints. Establishing a higher minimum or a lower maximum therefore requires additional observations or justified restrictions on how people allocated their time.

When exact labour totals are unavailable, a probability sample of person-intervals (time intervals for individual people) can give an estimate that is unbiased over repeated samples, provided inclusion probabilities are known and positive and allocations are measured accurately. For each sampled person-interval, divide observed labour minus predicted labour by the interval's inclusion probability. Add these weighted differences to the total of predictions fixed before sampling. This estimator improves precision when the sum of weighted differences has lower variance over repeated samples than the estimator without predictions.

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Telemetry can therefore improve a labour estimate even when commit events do not uniquely determine labour. Sampling uncertainty describes how an estimate varies with the selected sample. An identification interval instead contains exactly the labour totals consistent with the specified observations and allocation constraints. Estimating an intervention effect additionally requires a causal study design.

Wright and Ziegler's Standard Coder approach [13] infers coding activity from commit timing and predicts the resulting labels from code-change features. Standard Coding Hours (SCH) can serve as a common reference index. Using SCH to measure person-hours requires calibration against independently observed labour. AI may alter both how human activity produces commits and how much work a code change represents. The SPACE framework places activity and efficiency within a broader account of developer productivity [7]. Mozannar et al. examine interaction time costs in AI-assisted programming [10].

Existing AI studies measure different outcomes in different populations. Peng et al. examine completion time among completers of a programming exercise [11]. Becker et al. study experienced maintainers in familiar repositories, comparing log completion times after adjustment for forecast times [2]. Cui et al. study weekly completed pull requests in organisational experiments [5], with assignment used as an instrument for adoption in their June 2025 manuscript [4, Section 4.1 and Table 3]. Their estimates concern different populations, interventions and outcomes, so they cannot be substituted for one another as estimates of the same quantity. METR's subsequent account of selection and concurrent-agent measurement difficulties further motivates separate labour and elapsed-time records [3]. The calculations here use no data from these studies.

Sections 2 and 3 explain why commits alone need not determine person-hours and define labour, acceptance time and retained output. Section 4 derives the minimum and maximum possible labour totals, and Section 5 uses sampled labour to correct fixed telemetry predictions. Section 6 sets out how to validate these measurements before comparing interventions. The proofs apply standard optimisation and sampling results to the stated constraints on allocating people's time. Appendix A gives a shortest-path solution for one task–person–role combination.

2. Why event telemetry alone need not identify labour

Let a minute-indexed hidden variable $Z_t \in \{0,1\}$ indicate an activity state, and let $C_t \in \{0,1\}$ record a commit event. A fitted event model specifies how activity changes from minute to minute and how activity produces commits. Interpreting $Z_t = 1$ as "coding" requires validation against independently measured labour. Even before that validation, different probability distributions of activity can produce the same distribution of commit events.

Proposition 2.1 (Failure of unrestricted identification)**.** *Suppose the states Z_t are independent Bernoulli variables with parameter $\pi \in (0,1)$, and, independently across minutes and of the states, B_t are Bernoulli variables with parameter $c \in (0,1)$. Define $C_t = Z_t B_t$. The joint distribution of all commit observations depends on (π, c) only through πc. In particular, $(\pi, c) = (1/2, 1/5)$ and $(1/4, 2/5)$ give identical commit distributions but different expected active times.*

Proof. Each C_t is Bernoulli with parameter $\theta = \pi c$, and the C_t are independent. For every $n \geq 1$ and every binary string $(e_1, \ldots, e_n)$,

$$\mathbb{P}(C_1 = e_1, \ldots, C_n = e_n) = \theta^{\sum_t e_t}(1-\theta)^{n-\sum_t e_t}. \tag{1}$$

Both parameter pairs give $\theta = 1/10$. In contrast,

$$\mathbb{E}\left[\sum_{t=1}^{n} Z_t\right] = n\pi, \tag{2}$$

which is $n/2$ minutes in the first model and $n/4$ in the second. Thus identical commit distributions can correspond to different expected numbers of active minutes. □

This construction belongs to the independent-state subfamily of a two-state hidden Markov model: the next activity state has the same distribution for either current state. The example shows that different parameter values in the unrestricted model family can give the same distribution of commits. Models with persistent activity or other restrictions may still be identifiable. Restrictions on the probability model for activity, informative code-change features or independent activity observations may distinguish models that commit timing alone cannot distinguish. A claim that the parameters are identifiable must state which restrictions or observations make that possible. Under AI assistance, commits can continue while human attention is elsewhere, so the mechanism that produces commits may itself change.

Agreement between models does not establish that their predictions agree with independently measured person-hours. Measured interval totals instead constrain how much labour occurred during a specified period.

3. Task accounting with explicit boundaries

3.1. **The task, the assigned intervention and the person-hour.** Enrol task i at time s_i, when its requested behaviour, eligibility and acceptance criteria are recorded. Give the task an identity tied to its requirement, so changing its commits or pull requests does not change which task is counted as output. Before observing outcomes, define how split tasks, merged tasks and scope changes will be handled. A parent requirement can retain its original identity while children record implementation activity. Counting both the parent and its children as delivered output would count the same work twice. Changes affecting the stated requirement or acceptance criteria require a versioned record and a stated analysis rule.

Before interventions are compared, the task-accounting rules must specify what counts as one task, which task receives acceptance credit and when follow-up starts. Consider separately accepted components regrouped into a parent task that is accepted only after all its components have been accepted. Even when component reference scores add to the parent's score, regrouping can change how much output is credited by a deadline. The comparison therefore depends on how tasks are grouped and accepted as well as how they are scored.

An *AI intervention* specifies available tools and model versions, eligible tasks, permitted autonomous actions, review responsibilities and release requirements. A comparison of interventions must specify what is assigned and record actual use separately, since difficult tasks may elicit more use and people may decline an assigned tool.

Let $a_{ijr}(t)$ be the fraction of person j's time allocated to task i in role r at calendar time t. With finite sets of tasks, people and roles, measurable allocation functions define cumulative labour through task age $u \geq 0$, measured from enrolment, by

$$a_{ijr}(t) \geq 0, \qquad \sum_{i,r} a_{ijr}(t) \leq 1, \qquad H_i(u) = \sum_{j,r} \int_{s_i}^{s_i+u} a_{ijr}(t)\, dt. \tag{3}$$

With time in hours, $H_i(u)$ has units of person-hours. Include every role contributing labour to the task, from specification and implementation through review, testing and repair. Prompting and integration also consume person-hours. Pair work can contribute two person-hours in one elapsed hour. For one person, two concurrent sessions cannot each receive a full hour of attention. State whether attended waiting belongs in the labour total. Unattended execution is recorded as agent resource use.

To measure the task labour defined in Equation (3), specify how shared review and platform work are assigned to tasks. Use an overhead category for work that the observations cannot assign to particular tasks. Record agent instance-hours by resource class, tokens and charges

separately from person-hours. Combining these quantities requires an explicit economic conversion and changes the question from labour to cost.

Let T_i be elapsed time from enrolment to first acceptance. Starting this clock at enrolment includes queueing and any reasoning and specification between enrolment and the first commit. If the task is permanently abandoned before acceptance, put $T_i = \infty$. Abandonment is an outcome, whereas loss of observation is censoring. Record deployment separately whenever acceptance and deployment differ.

3.2. **A frozen reference and a follow-up assessment.** Assign a nonnegative reference score S_i^0 using a declared scoring-rule version and task representation. It may count tasks or functional units, or use frozen SCH weights. Retained output has the same unit as this score. Productivity divides that output by person-hours. The superscript 0 denotes the reference held fixed across interventions. A score expressed in person-hours needs calibration to independently measured labour for a specified reference population, included roles, quality criterion and stopping rule. A predictor trained on imputed activity states learns to predict those states; its accuracy as a measure of person-hours still requires independent validation.

For a causal comparison with fixed task requirements, requirements recorded at intake are preferable to final code differences: AI can change the final code. A retrospective code score can still be reported, with its dependence on the implementation made explicit. A nonlinear scoring rule need not be additive, so splitting or squashing a change can alter its score. Score the same recorded representation of each task once, or use a requirement rubric whose component scores add to its total. A failed attempt, retry or revert must not generate another unit of retained output.

Choose an acceptance deadline at task age $\tau > 0$ and a follow-up duration $\Delta \geq 0$. For accepted tasks, let $R_i(\Delta) \in [0, 1]$ be the assessed fraction of the accepted requirement still meeting the rubric at time $s_i + T_i + \Delta$. Retained reference output is

$$Q_i(\tau, \Delta) = \begin{cases} S_i^0 R_i(\Delta), & T_i \leq \tau, \\ 0, & T_i > \tau. \end{cases} \tag{4}$$

The rubric must distinguish an intentional replacement from a defect-driven loss and explain fractional scores. Surviving lines, successful tests and absence of recorded incidents each provide observations of quality, while leaving complete correctness unestablished. Validation must assess agreement between reviewers and document what the tests and rubric can detect.

For a finite intake cohort with positive total labour, define

$$P_H(\tau, \Delta) = \frac{\sum_i Q_i(\tau, \Delta)}{\sum_i H_i(\tau + \Delta)}. \tag{5}$$

The denominator is total cohort labour through the same task age $\tau + \Delta$ for every enrolment. It includes labour on rejected, unfinished and abandoned tasks, together with work done after acceptance that is attributable to the task. Early acceptance leaves more time for repair within this labour window than late acceptance. Stopping repair accounting exactly Δ after each acceptance would define a different denominator. State when labour recording ends and follow every task for $\tau + \Delta$ to measure both the retained output and labour defined here.

Tasks with zero human labour can contribute to the cohort's retained output. The productivity ratio is defined only when total cohort labour is positive. Report retained output and total person-hours separately.

3.3. **The observations and the three clocks.** Acceptance, retained output and labour use different timing rules: acceptance by the deadline τ, retention at task age $T_i + \Delta$, and labour accumulated through task age $\tau + \Delta$. A complete task history supplies the observations needed under each rule. Recording acceptance alone does not establish how much output survives or how much repair work follows. A record *point-identifies* realised

labour when every way of allocating time that satisfies the record and the model's constraints gives the same total. Proposition 2.1 instead asks whether the probability distribution of observed events uniquely determines a population parameter. The next section derives the smallest and largest labour totals consistent with interval observations.

4. Attainable labour ranges from interval observations

Knowing how many person-hours occurred during an interval need not tell us when within that interval people worked. We seek every labour total that could have occurred within the specified target windows while satisfying the observations and the allocation constraint in (3). These totals form the *identified range*: for each total, at least one way of allocating people's time satisfies all the observations and constraints. No allocation satisfying them gives a total outside this range. This definition follows the general partial-identification perspective [12, Section 1]. The following proofs derive these totals from the stated observation bounds and allocation constraints.

4.1. Using the person's capacity.

One person can contribute at most one person-hour in an elapsed hour. Knowing who supplied an interval total therefore limits how much of that labour could have occurred before or after a specified time. A total across several people must be compared with their combined available time.

Proposition 4.1 (Attainable contribution from one interval)**.** *Let $\ell < u$, $c \in [\ell, u]$ and $v \in [0, u-\ell]$. Among measurable allocations $a : [\ell, u] \to [0,1]$ with $\int_\ell^u a(t)\,dt = v$, the attainable values of $x = \int_\ell^c a(t)\,dt$ are exactly*

$$[\max\{0, v-(u-c)\},\ \min\{v, c-\ell\}]\,. \tag{6}$$

If $v < 0$ or $v > u-\ell$, no such allocation exists.

Proof. Nonnegativity and capacity on the two pieces imply

$$0 \le x \le c-\ell, \qquad 0 \le v-x \le u-c.$$

Together, these inequalities give (6). Conversely, for any x in this interval, set $a = x/(c-\ell)$ before c and $a = (v-x)/(u-c)$ after c. Each value lies in $[0,1]$ and gives the required integral. Omit a zero-length piece when c is an endpoint: the corresponding numerator is then zero. Finally, integrating $0 \le a \le 1$ gives the necessary condition $0 \le v \le u-\ell$. □

For one person, 3/2 person-hours in a two-hour interval imply between 1/2 and 1 person-hours in its first hour. Nonnegativity alone would give $[0, 3/2]$. In this example, the bounds do not determine the first-hour total or exactly when the person worked. If $v = u-\ell$, the total uses all the person's available time, so the labour before any time within it equals the elapsed time from its start. For an interior subinterval $[b, c] \subseteq [\ell, u]$, applying the same construction to that subinterval and the time outside it gives every possible labour total in the subinterval: $[\max\{0, v-((u-\ell)-(c-b))\}, \min\{v, c-b\}]$.

Bounds calculated separately for each contribution can be added when each contribution is counted once. The summed lower bound is possible only if allocations achieving every component minimum can occur together; the same condition applies to the upper bound and component maxima. Disjoint intervals of one person, or intervals belonging to distinct people, can be combined when no other constraints link them. Overlapping observations or several tasks competing for the same person's time require a joint calculation. For example, one person spends one person-hour on each of two tasks during $[0, 2]$. Each task separately allows $[0, 1]$ person-hours before time 1. Together, the tasks require exactly one person-hour before time 1, because they use all the person's available time on $[0, 2]$. Adding the separately calculated bounds would give $[0, 2]$.

4.2. **Replacing allocation histories with interval totals.** To account for tasks sharing a person's time, call each task–person–role combination a stream. Fix finite sets of tasks $\mathcal{I}$, people $\mathcal{J}$ and roles $\mathcal{R}$, a cohort $\mathcal{C} \subseteq \mathcal{I}$, and a bounded calendar-time domain $[b_0, b_K]$ with $b_0 < b_K$ containing every observation interval and target window. An admissible history consists of measurable $a_{ijr} \geq 0$ satisfying $\sum_{i,r} a_{ijr}(t) \leq 1$ almost everywhere for every person j. The declared roster and rules for assigning time to tasks determine whose labour and which work are included. Whether these declarations cover the intended work requires measurement evidence. Work outside $\mathcal{C}$ can be recorded separately when it uses the same people's time without contributing to cohort labour.

For observations $\nu = 1, \ldots, M$, specify an interval $[\ell_\nu, u_\nu]$ within this domain, a set $G_\nu \subseteq \mathcal{I} \times \mathcal{J} \times \mathcal{R}$ of attributed streams, and finite labour bounds $v_\nu^- \leq v_\nu^+$, in person-hours. Impose

$$v_\nu^- \leq \sum_{(i,j,r) \in G_\nu} \int_{\ell_\nu}^{u_\nu} a_{ijr}(t)\, dt \leq v_\nu^+. \tag{7}$$

Exact interval totals use $v_\nu^- = v_\nu^+$. Justified limits on rounding or measurement error may instead give different lower and upper bounds. A Horvitz–Thompson estimate varies with the selected sample; that sampling uncertainty requires a separate analysis before the estimate can justify either inequality.

Form the sorted distinct endpoints $b_0 < \cdots < b_K$ from the domain boundaries, every observation endpoint and every target boundary. Call the intervals between successive endpoints cells. For the cohort denominator, target boundaries include each cohort task's enrolment s_i and the fixed end of its labour observation window $c_i = s_i + \tau + \Delta$. Translate task ages into calendar time before partitioning, so that simultaneous work by the same person shares one capacity constraint. Keep cells outside a target window whenever an observation spans them: some of the observed labour may have occurred outside the window. Put $d_k = b_k - b_{k-1} > 0$ and let x_{ijrk} denote the labour allocated to stream (i, j, r) in cell k.

Proposition 4.2 (Exact cell reduction and attainable labour range)**.** *Under the preceding model, define the finite-dimensional feasible set $\mathcal{P}$ by*

$$x_{ijrk} \geq 0, \qquad \sum_{i,r} x_{ijrk} \leq d_k \qquad \text{for every } j, k, \tag{8}$$

$$v_\nu^- \leq \sum_{(i,j,r) \in G_\nu} \sum_{k:\, [b_{k-1}, b_k] \subseteq [\ell_\nu, u_\nu]} x_{ijrk} \leq v_\nu^+ \qquad \text{for every } \nu. \tag{9}$$

Fix real weights w_{ijrk} and define the target for a continuous-time allocation history by multiplying each cell integral by w_{ijrk} and summing. For its finite representation $h(x) = \sum_{i,j,r,k} w_{ijrk} x_{ijrk}$, the attainable target values over admissible histories equal $h(\mathcal{P})$. In particular, the fixed-horizon cohort denominator uses $w_{ijrk} = 1$ exactly when $i \in \mathcal{C}$ and $[b_{k-1}, b_k] \subseteq [s_i, c_i]$, and zero otherwise. If $\mathcal{P}$ is nonempty, its target range is the closed interval

$$[H_{\min}, H_{\max}] = \left[\min_{x \in \mathcal{P}} h(x),\ \max_{x \in \mathcal{P}} h(x) \right]. \tag{10}$$

Both endpoints and every value between them are attained. If $\mathcal{P}$ is empty, the observations and capacity assumptions are inconsistent.

Proof. Integrating any admissible history over each cell gives nonnegative cell labour totals. Integrating each person's allocation limit gives (8). Splitting each observation at the cell boundaries gives (9). The weighted target total is unchanged because its weights are constant on cells.

Conversely, for $x \in \mathcal{P}$ set $a_{ijr}(t) = x_{ijrk}/d_k$ in the interior of cell k, and set the finitely many endpoint values to zero. These measurable allocations are nonnegative and have sum at most one for each person. They reproduce every cell labour total, so all observation

bounds remain satisfied and the target value is unchanged. Thus the original allocation functions and the finite feasible set give exactly the same possible target values.

The constraints define a closed convex set. Each coordinate is between zero and d_k, so a nonempty $\mathcal{P}$ is compact and a linear functional attains its extrema. Convex combinations of minimising and maximising points attain every intermediate value. The reconstruction just given supplies an admissible history for each of them. □

The minimum and maximum target totals can be computed by linear programmes, each with $|\mathcal{I}||\mathcal{J}||\mathcal{R}|K$ variables, $|\mathcal{J}|K$ shared-capacity constraints and $2M$ observation inequalities, in addition to nonnegativity. For every assignment of cell labour totals satisfying the linear constraints, the proof constructs time allocations that satisfy the original observations and give the same target total. A known period of unavailability can be included by adding its boundaries to the partition and setting that person's cell labour totals to zero within it. This exact correspondence depends on allowing measurable fractional allocations. Minimum session lengths or restrictions on task interruption would limit which time allocations are permitted. Smoothness constraints and switching costs would also require a separate analysis.

For the cohort-denominator target, let $Q = \sum_{i \in \mathcal{C}} Q_i(\tau, \Delta)$ be retained output known exactly. With $Q \geq 0$ and $H_{\min} > 0$, dividing this output by the largest and smallest possible labour totals gives the identified productivity interval $[Q/H_{\max}, Q/H_{\min}]$. Justified quality bounds $0 \leq Q_- \leq Q \leq Q_+$ give an interval $[Q_-/H_{\max}, Q_+/H_{\min}]$ containing every feasible output-to-labour ratio. It is the exact range only when every ratio in it can be produced by an output total and labour total that are jointly permitted. It is sufficient for every pair in $[Q_-, Q_+] \times [H_{\min}, H_{\max}]$ to be permitted. A lower labour bound of zero cannot be used as a divisor. The acceptance clock and retention assessment in (4) remain fixed throughout the allocation optimisation.

5. Estimating labour from sampled allocations

The preceding bounds assume exact interval labour totals or justified lower and upper bounds on them. A probability sample instead gives an estimate that varies with the selected person-intervals. Treating that estimate as an exact interval total can rule out the true work allocation because of sampling error. The difference estimator below corrects telemetry predictions using sampled labour. Under explicit sampling and measurement assumptions, its average over repeated samples equals the true labour total.

Activity sampling estimates labour from selected person-intervals. The sampling frame must list person-intervals covering all target work, including off-keyboard reasoning, review and coordination. Short diaries and task records can show how much of each selected interval a person spent on the target work. Independent assessment of a subsample measures disagreement over these allocations. Nonresponse, inaccurate recall and work outside the frame require separate investigation.

5.1. **Frozen predictions as auxiliary information.** The difference estimator starts with a fixed prediction of total labour and corrects it using a probability sample of allocation observations. For each sampled person-interval, subtract predicted labour from observed labour and divide by the interval's inclusion probability. Under the conditions below, the sum of these weighted residuals averages over repeated samples to the true total minus the prediction total. Denis and Haddou give this fixed-prediction estimator and its variance under the sampling design explicitly [6, Section 2.3, equations (2)–(4)]. Prediction-powered inference also identifies its estimator of a mean as a difference estimator and considers finite-population extensions [1, Section 1.4 and Appendix B of the accompanying preprint].

Partition each person's target observation period into non-overlapping intervals. Let $U = \{1, \ldots, M\}$, $M \geq 1$, be the complete, fixed frame of these person-intervals. For interval m, let $d_m > 0$ be its length in hours and $a_m \in [0, 1]$ its average allocation to the target work. Its labour is $y_m = d_m a_m$ person-hours, and the population total is $H = \sum_U y_m$. For

every frame unit, retain a prediction f_m in person-hours. A raw SCH value requires a stated conversion to labour units before it can serve as f_m. Each unit's prediction must remain the same whichever sample is drawn. Training on an independent earlier population can satisfy this condition: hold all predictions fixed, then sample allocations to correct the predicted total.

Proposition 5.1 (Fixed-prediction difference estimator)**.** *For a probability sample $s \subseteq U$ with known first-order inclusion probabilities $\pi_m > 0$, set $r_m = y_m - f_m$ and*

$$\widehat{H}_{\text{diff}} = \sum_{m \in U} f_m + \sum_{m \in s} \frac{y_m - f_m}{\pi_m}. \tag{11}$$

If y_m and f_m are fixed under the design and every sampled labour value is observed accurately, then $\mathbb{E}_s[\widehat{H}_{\text{diff}}] = H$, where $\mathbb{E}_s$ averages over the sampling design.

Proof. Write $I_m = \mathbf{1}\{m \in s\}$. Linearity and $\mathbb{E}_s[I_m] = \pi_m$ give

$$\mathbb{E}_s[\widehat{H}_{\text{diff}}] = \sum_{U} f_m + \sum_{U} \frac{\mathbb{E}_s[I_m]}{\pi_m}(y_m - f_m) = \sum_{U} y_m.$$

□

The residual correction makes the estimate design-unbiased even when the fixed prediction total differs from H. Taking $f_m = 0$ gives the Horvitz–Thompson (HT) estimator [8],

$$\widehat{H}_{\text{HT}} = \sum_{m \in s} \frac{d_m a_m}{\pi_m}. \tag{12}$$

For every frame interval, record one of three states: selected with an observed allocation (including zero), selected with no recorded allocation value (nonresponse), or unselected. The proof concerns true allocation values. Unbiasedness in the presence of measurement error requires additional assumptions on that error.

5.2. **Precision depends on residuals and the sampling design.** Let $\pi_{mk} = \mathbb{E}_s[I_m I_k]$ be the probability that both frame units are selected, including $\pi_{mm} = \pi_m$. The exact variance over repeated samples is

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \sum_{m,k \in U} \frac{\pi_{mk} - \pi_m \pi_k}{\pi_m \pi_k} r_m r_k. \tag{13}$$

Expanding the variance of $\sum_U I_m r_m / \pi_m$ gives this expression. The calculation depends on how often pairs of units are selected together, through the covariances of their inclusion indicators. The variance formula in (13) also uses residuals for unsampled units, which are generally unknown. To estimate it, retain the actual sampling design and satisfy the chosen variance estimator's conditions on joint inclusion probabilities.

For simple random sampling without replacement from $M \geq 2$ units, let the fixed sample size satisfy $1 \leq n \leq M$. Every size-n subset has the same probability and $\pi_m = n/M$. Write

$$S_r^2 = \frac{1}{M-1} \sum_{U} (r_m - \bar{r}_U)^2, \qquad S_y^2 = \frac{1}{M-1} \sum_{U} (y_m - \bar{y}_U)^2.$$

The inclusion covariances in (13) give

$$\operatorname{Var}_s(\widehat{H}_{\text{diff}}) = \frac{M^2}{n}\left(1 - \frac{n}{M}\right) S_r^2, \qquad \operatorname{Var}_s(\widehat{H}_{\text{HT}}) = \frac{M^2}{n}\left(1 - \frac{n}{M}\right) S_y^2. \tag{14}$$

For $2 \leq n \leq M$, an unbiased estimator of the design variance is

$$\widehat{V}_{\text{diff}} = \frac{M^2}{n}\left(1 - \frac{n}{M}\right) s_r^2, \qquad s_r^2 = \frac{1}{n-1} \sum_{m \in s} (r_m - \bar{r}_s)^2. \tag{15}$$

Indeed, $s_r^2 = [n(n-1)]^{-1} \sum_{m<k,\, m,k \in s} (r_m - r_k)^2$. Each pair has inclusion probability $n(n-1)/[M(M-1)]$, so $\mathbb{E}_s[s_r^2] = S_r^2$. Selecting one unit from a population containing more than one unit does not supply this variance estimate. Measuring every unit gives zero sampling variance, including in a single-unit population. Measurement error remains a separate question.

For $n < M$, the difference estimator has lower sampling variance than HT exactly when $S_r^2 < S_y^2$: the residuals must have lower population variance than the person-interval labour totals. Consider $M = 4$, $n = 2$, and $y = (0, 2, 4, 6)$. The true total is 12, and HT variance is $80/3$. Frozen predictions $f = (1, 3, 3, 5)$ yield variance $16/3$, whereas $f = (6, 4, 2, 0)$ yield $320/3$. Enumerating all six samples reproduces these variances and confirms that each estimator averages to the true total. It also verifies that (15) averages to each corresponding variance. The second predictor gives four times the HT variance. The difference estimator remains design-unbiased with either predictor. These finite checks establish the calculations for these two predictors. A workplace study must determine whether its code-derived predictions give a residual total with lower sampling variance than the labour estimate without predictions.

5.3. **Failure cases and implementation checks.** The proof above does not cover predictions fitted to the sample used for correction. For $M = 2$, $n = 1$ and $y = (0, 2)$, suppose a fitted rule memorises the selected outcome and predicts zero for the other unit. Its observed residual is always zero. The difference estimate is nevertheless 0 or 2, each with probability $1/2$, giving expectation 1 rather than $H = 2$. A method trained on the sampled units needs a separate unbiasedness argument under the actual finite-population sampling design.

The estimate can also be negative even though true labour is nonnegative. With $y = (0, 0)$, $f = (10, 0)$ and one uniformly selected unit, (11) is -10 or 10 with equal probability. Clipping at zero changes its expectation from 0 to 5. Preserve the signed estimate when reporting this estimator and investigate an implausible result. Dividing retained output by an unbiased labour estimate also need not yield an unbiased productivity estimate.

The accompanying implementation reports estimates with their calculated sign, preserves observed zeros, and records nonresponse separately from unselected units. It rejects declarations with missing required information, selected intervals without allocation measurements, and inconsistent inputs. For a general declared design it reports point estimates. For simple random sampling it also reports variance and standard error under the conditions above. These checks establish whether supplied records satisfy the input rules. Establishing that the frame covers the work, allocations are measured accurately and units were sampled by the declared method requires evidence from collection.

Exhaustive finite tests check the examples above, input validation, complete censuses and single-unit populations. A confidence interval requires a separate argument that its stated coverage holds under the sampling design.

6. Validating the measurements

An empirical study should first test whether the proposed observations measure person-hours accurately and cover all the included work. Before collection, register the task scope and attribution rules that determine which labour is counted. Fix the sampling frame, reference score, acceptance horizon and retention rubric for the same population. Measure allocation disagreement, nonresponse, follow-up completeness and collection burden by intervention, role and task class. Use these observations to choose how many person-intervals to sample and how precisely a later study should estimate intervention differences.

For interval bounds, use independently assessed totals with justified limits on measurement error. For prespecified target windows, compare the sums of separately calculated lower and upper labour bounds with the joint minimum and maximum. This shows whether accounting

jointly for each person's available time raises the minimum or lowers the maximum. For sampled labour, compare the variance over repeated samples with and without predictions frozen before sampling. Retain the inclusion probabilities and joint sampling-design information needed to calculate uncertainty.

6.1. **Comparing predictions with sampled labour.** A sampled labour estimate can differ from true labour, so its difference from a prediction includes sampling error. Compare frozen telemetry predictions by assigned intervention, role and task class on held-out teams or periods, while accounting for that sampling error. Correlation between group averages does not establish whether predictions systematically overstate or understate labour, or how far individual predictions are from true labour. The following calculations separate prediction error against true labour from sampling variance.

Treat each task's labour H in a finite validation population as fixed. Let f be its prediction, frozen before validation sampling, and suppose $\mathbb{E}_s[\widehat{H}] = H$ with finite design variance. Then

$$\mathbb{E}_s[f - \widehat{H}] = f - H,$$
$$\mathbb{E}_s[(f - \widehat{H})^2] = (f - H)^2 + \mathrm{Var}_s(\widehat{H}).$$

Expand the square: the cross term vanishes because $\mathbb{E}_s[\widehat{H} - H] = 0$. These identities hold for each task under its stated sampling design. Aggregate uncertainty additionally depends on covariances between tasks' estimates.

For a second prediction g frozen before the same validation sample, subtraction cancels the common sampling variance:

$$\mathbb{E}_s[(f - \widehat{H})^2 - (g - \widehat{H})^2] = (f - H)^2 - (g - H)^2.$$

Over repeated validation samples, the average difference between the two squared prediction errors equals their difference calculated against true labour. Estimating either predictor's squared error against true labour, or quantifying sampling uncertainty in their comparison, requires further sampling-design information. Absolute error requires a separate analysis.

For example, eight one-hour person-intervals have allocations $(1, 1, 1, 1, 0, 0, 0, 0)$. Sample one uniformly, with inclusion probability $1/8$. True labour is $H = 4$, whereas its HT estimate is 0 or 8, each with probability $1/2$. A perfect prediction $f = 4$ has apparent absolute error 4 against the sampled estimate. The interval $[3.5, 4.5]$ always contains true labour but never contains the sampled estimate. The signed difference between prediction and sampled estimate remains unbiased for the prediction's error against true labour. Absolute error and interval coverage calculated against $\widehat{H}$ measure agreement with the sampled estimate, including its sampling error.

To assess accuracy against true labour, select a validation subset of tasks, record allocations for every person-interval in each task's target window, and adjudicate the records independently. Report remaining measurement error. Alternatively, specify a measurement model under which the observations uniquely determine the chosen measure of prediction accuracy. Retain inclusion probabilities and joint sampling-design information for uncertainty calculations.

Validation should also test whether task splitting, code-change squashing or changes to the task representation alter the reference score. Report reviewer disagreement and follow-up completeness alongside prediction error. Recalculate intervention comparisons using plausible amounts of missing labour and disputed retention scores. If these changes reverse which intervention ranks higher, choosing an intervention requires resolving or accepting the measurement uncertainty.

6.2. **Conditions for an intervention comparison.** A subsequent causal study must define whether it concerns a fixed set of tasks or an organisation's eligible incoming work, which can change under the intervention. Freeze task eligibility and the treatment of pre-assignment work for a fixed-workload comparison. Record postponed, rejected and

abandoned requests. An organisation-wide adoption comparison instead needs a common calendar window and records of intake and work in progress. State which population each comparison concerns and how its observations are weighted.

An intervention assigned to one team can affect another team through shared reviewers, agent capacity or reusable components. The assignment design and analysis must account for these effects between teams. Analyse tasks by assigned intervention and calculate uncertainty using the units assigned to interventions, with additional allowance for labour and quality subsampling. Recording more commits does not increase the number of independent assignments.

Use the same retention rubric and follow-up procedure across intervention groups. Duplicate assessment of a subset measures disagreement. Record failures of reviewer blinding and the timing of feedback, since assessment can trigger repair. Count that repair within the declared labour window. Estimating retained output for all tasks from an assessed subset requires known selection probabilities and assumptions addressing any effect of assessment on subsequent work. Report severe defects separately because a retained fraction cannot represent every security or reliability consequence.

Complete follow-up through $\tau + \Delta$ as required by Section 3. Distinguish permanent abandonment from loss of observation. If follow-up is incomplete, state the assumptions or bounds used to handle missing outcomes. Report labour, acceptance and retained output together, since reduced labour may accompany more abandonment. Report retained output and positive total labour alongside their productivity ratio, and account for uncertainty in both quantities and their dependence. Prespecify practically useful improvement thresholds and any maximum acceptable reduction in quality (the noninferiority margin). Failure to detect a quality reduction does not establish that the loss is within that limit.

7. Scope and next observations

Different distributions of human activity can produce identical commit distributions. Interval totals or justified bounds, together with each person's available time, restrict how much work could have occurred during a specified period. Splitting time at every observation endpoint and target-window boundary gives a finite optimisation problem with exactly the same possible labour totals. Adding separately calculated interval bounds can include totals that would require someone to work beyond their available time. The summed lower and upper bounds are exact only if work allocations achieving the respective component bounds can occur together.

Probability sampling permits design-unbiased estimation when the frame covers the target work, inclusion probabilities are known and positive, and sampled allocations are measured accurately. A fixed telemetry predictor improves precision only when estimating its total residual has lower sampling variance than estimating labour without predictions. Diaries and task records can supply the observations. Whether they capture reasoning and coordination and accurately record how people divide their time requires empirical evidence. The unbiasedness and precision results depend on the stated measurement and sampling assumptions. Establishing a causal benefit from AI adoption requires a further comparison of interventions.

The next empirical test is whether records tied to each task can account for person-hours from every included role and independently assessed retained output through the required follow-up. Disagreement over recorded allocations, missing records and collection burden will help determine how much measurement is practicable before an intervention trial is designed. That trial must also choose whether to assess a fixed workload or an organisation-wide adoption that can change incoming requirements. The results specify which labour totals are consistent with the observations, when sampling gives an unbiased estimate and when fixed predictions reduce its variance.

Reproducibility

The repository at https://github.com/emmett08/engineering-work-observatory/tree/main/paper contains the LaTeX source, exact rational interval checker and finite sampling checks. The analytical proofs establish the general results. The executable checks test the calculations on constructed data; workplace measurements require empirical validation. For feasible inputs within its resource limits, the application solves the joint shared-capacity programme in exact rational arithmetic and returns work allocations achieving the minimum and maximum target totals. Its existing cohort reports use separately calculated floating-point bounds without guaranteeing that the rounded interval contains the interval obtained with exact arithmetic. For version-2 inputs, the estimator checks that each person's declared intervals cover their observation window without gaps or overlaps. Design-variance estimates are available for simple random sampling when the sample size permits. Establishing coverage of actual work, allocation accuracy and compliance with a sampling design requires evidence from collection.

The unchanged historical package retains Lean proofs of the finite observation-law counterexample and inverse-probability-weighting algebra. The attainable interval bounds, joint cell reduction, difference-estimator variance and sampling-validation arguments developed here have the analytical proofs and computational checks described above. They lie outside the historical proof receipt's coverage.

Appendix A. A shortest-path solution for one allocation stream

For one allocation stream $0 \le a \le 1$, the interval observations of Section 4 can be represented by cumulative labour at their endpoints. Using the sorted endpoints and cell lengths defined there, set $z_k = \int_{b_0}^{b_k} a(t)\,dt$. Write an inequality $z_q - z_p \le w$ as a directed edge $p \to q$ of weight w. Adjacent endpoints receive edges $k-1 \to k$ of weight d_k and $k \to k-1$ of weight zero. Observation bounds on $[b_p, b_q]$ add edges $p \to q$ of weight v^+ and $q \to p$ of weight $-v^-$. An exact observed labour total v therefore gives edge weights v and $-v$. This is the standard difference-constraints representation [9, Slides 6–11].

Proposition A.1 (Attainable cumulative bounds from graph distances)**.** *The scalar observations admit an allocation if and only if this graph has no directed cycle of negative total weight. When it is feasible, let $D(p,q)$ be the shortest-path weight from p to q. For any endpoint pair $p \le q$, the identified range of $\int_{b_p}^{b_q} a(t)\,dt$ is*

$$[-D(q,p),\ D(p,q)]. \tag{16}$$

Proof. Summing edge inequalities around a cycle gives 0 at the left, so a negative cycle is inconsistent. Conversely, adjacent edges make every vertex reachable from every other. In the absence of negative cycles, all shortest-path weights are finite and $D(r,r) = 0$. For any starting vertex r, the values $z_k = D(r,k)$ satisfy each edge inequality because $D(r,v) \le D(r,u) + w$ for an edge $u \to v$ of weight w. Subtracting z_0 makes initial cumulative labour zero without changing differences. The inequalities for adjacent endpoints give cell labour totals in $[0, d_k]$. Allocations constant within each cell can produce these totals and satisfy the observations.

Summing inequalities along paths yields $z_q - z_p \le D(p,q)$ and $z_q - z_p \ge -D(q,p)$. Using p as the starting vertex gives an allocation achieving the upper bound; using q gives an allocation achieving the lower bound. Convex combinations of these allocations give every labour total between those bounds. □

The reference checker uses exact rational arithmetic and an all-pairs shortest-path calculation. Given indexed, sorted endpoints, graph construction and this calculation use $O((K+1)^3 + M)$ arithmetic operations and $O((K+1)^2 + M)$ storage. Bit costs depend on the encoded rational inputs. The checker compares the bounds and the allocations attaining

them against an independent enumeration of finite allocations, including overlapping interval observations and inconsistent inputs. These constructed examples check the calculation against the proved result. Empirical measurements are still needed to validate the observation process. Several task streams sharing a person's capacity require the joint formulation in Proposition 4.2, because the scalar graph represents only one allocation.

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